

National University of Computer and Emerging Sciences

Karachi Campus

Data Science for Software Engineering (SE-4096)
Semester: Spring-2026
Date: Thursday, 19th Feb, 2026
Course Instructor(s) Dr Muhammad Yaseen Khan

Quiz 01
Total Time: 15 minutes
Total Points: 4
Total Questions: 02

Student Name

Roll Number

Section

Signature

Carefully read the following scenario and answer the questions below.

You are the Lead Software Test Engineer in a software company developing web and mobile applications. Recently, releases have been delayed due to missed defects, unstable builds, and uneven tester productivity. Some testers consistently execute fewer test cases and report fewer defects than others, affecting overall software quality.

Testing activities and results are recorded in repositories hosted on GitHub, along with issue trackers and CI/CD logs that store defect reports, build outcomes, execution times, and contributor information. You decide to apply Data Science for Software Engineering practices to analyze this data and generate productivity reports to support decisions such as workload redistribution, training, or probation for consistently low-performing testers.

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CLO 1: Students should be able to articulate a comprehensive understanding of Data Science, encompassing its fundamental concepts, methodologies, and practical applications in the Software Engineering discipline.

Question 1. Explain why applying Data Science for Software Engineering is necessary in this scenario and why testing-related repository data is important for evaluating tester productivity and improving software quality. [2 marks]

CLO 2: Students should be able to design and implement data pre-processing pipelines to clean, transform, and prepare raw data for analysis, ensuring data quality and integrity.

Question 2. Explain how you would collect, preprocess, and analyze the testing data to compute productivity metrics and identify underperforming testers. [2 marks]